Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering
Department of Computer Engineering(Regional Language)

Title: Dynamic Marathi Sign Language Ecosystem via Agentic AI

Subtitle: A Bi-Directional, Cloud-Native Accessibility Platform

Project Domain: Computer Vision, Artificial Intelligence, and Full-Stack Engineering

Presented by:

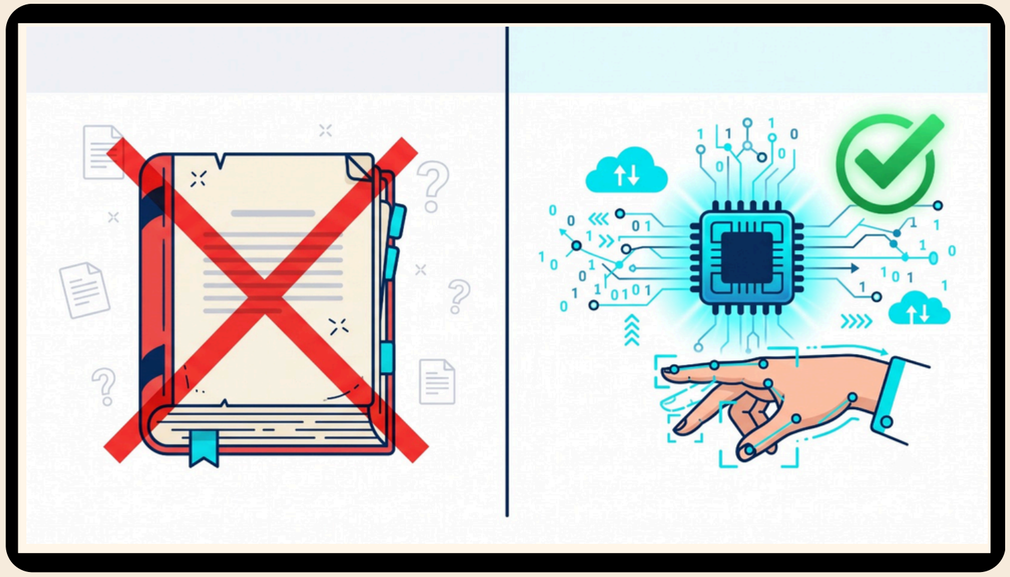
Adwait Kamble
Yadnesh Patil
Abhay Bormale
Atharva Zope

Under the guidance of Prof. Anandkumar Birajdar

Introduction & Problem Statement

Introduction:
Sign language is the primary mode of communication for the deaf and speech-impaired community. However, communication barriers arise because most hearing individuals do not understand sign language. While significant technological advancements have been made for American Sign Language (ASL) and Indian Sign Language (ISL), there is still a lack of intelligent and real-time communication systems specifically designed for Marathi Sign Language.

Problem Statement:
Existing sign language recognition systems are primarily static, rely on predefined dictionary mappings, and lack the ability to understand real-world conversational context.
They also face challenges such as varying lighting conditions, complex backgrounds, camera movement, and dynamic hand gestures.



Proposed Solution:
We propose an autonomous, bi-directional communication platform that translates real-time Marathi Sign Language into grammatically correct Marathi text and speech using YOLOv8 and Artificial Intelligence.
The system also enables hearing individuals to respond using text, which is converted into visual sign language through Text-to-Sign avatars.

Literature Review

2026

Paper Title & Reference	Year / Journal	Specific Methodology Referred / Adapted	Our Project Outcome & Implementation	Key Notes
"Indian Sign Language Recognition Using Canny Edge Detection" (IJATCSE)	2021 / IJATCSE	Extracting hand shape boundaries using traditional Canny Edge Detection for computer vision.	Found edge detection highly sensitive to background noise; abandoned traditional edge mapping and adopted YOLOv8m deep-learning bounding-box localization for robust tracking under varied lighting.	Shows limitations of classical edge methods for live, noisy scenes; motivated switch to deep object-detection approach.
"AI-Driven Indian Sign Language Recognition Using Hybrid CNN-BiLSTM Architecture for Divyangjan" (S. Nithyanandh)	2024/2025 / IJESE	Hybrid deep-learning using CNNs (and BiLSTM) for spatial feature extraction and temporal modelling for real-time recognition.	Validated moving from basic math filters to deep CNN backbone; implemented CSPDarknet53 inside YOLOv8 to capture complex, non-linear finger shapes dynamically.	Demonstrates benefit of CNN feature extractors and temporal modules for complex gestures; influenced backbone selection.
"Indian Sign Language Detection for Real-Time Translation using Machine Learning"	(year not specified)	Designing standard ML pipelines optimized for live, real-time sign translation (focus on inference speed).	Implemented multi-threading: separated pyttsx3 audio from OpenCV video feed so YOLO maintains ~30 FPS on a standard laptop.	Emphasizes inference speed and engineering needed for practical, real-time translation systems.
"Marathi sign language recognition methodology using Canny's edge detection" (Sadhana)	2024/2025 / Sadhana	Static image processing and Canny edge detection targeted at regional Marathi signs.	Demonstrated limitations for visually similar signs (e.g., Ka vs. Kha); applied 3x Roboflow data augmentation (Rotation, Blur, Noise) so model learns geometric differences beyond static edges.	Reinforced that augmentations improve robustness when static edge cues are ambiguous.
"CNN-Powered Marathi Sign Dialect Acknowledgment and Translation" (S. Mangale, R. Kashid, A. Kamble, et al.)	2024 / IERJ	Convolutional Neural Networks for dialect acknowledgment and classification of Marathi sign variations.	Upgraded from image-classification CNN to object-detection pipeline; implemented strict Region of Interest (ROI) Guide Box to remove facial/background bias before feeding frames to neural network.	ROI constraint reduced non-hand bias and improved detection/classification accuracy in dialect-sensitive cases.

Literature Review

2026

Paper Title & Reference	Year / Journal	Specific Methodology Referred / Adapted	Our Project Outcome & Implementation	Key Notes
"Real-Time Multi-Sign Language Recognition and Translation"	May 2026 / Everant Journals	Evaluated a unified pipeline mapping spatial bounding-box detectors (YOLOv8) alongside NLP layers to address environment-dependent variance across regional Asian dialects.	Validated our YOLOv8m object-detection setup; confirmed that direct bounding-box spatial tracking is more stable than coordinate skeletons under variable camera quality for Devanagari characters.	Validates bounding-box tracking over skeletal coordinates for environments with variable camera qualities.
"Advancing Bangla Sign Language Detection through Dataset Creation, Model Comparison, and Deployment"	June 2026 / MECS Press	Baseline comparison of YOLOv8n (Nano) vs. YOLOv8m (Medium), evaluating memory footprint and real-time laptop inference metrics.	Supported decision to scale to YOLOv8m: Nano is faster but suffers severe classification drops with slight hand orientation twists, so Medium was chosen to capture character intricacies.	Justifies upgrading to YOLOv8m to prevent accuracy drops from complex hand orientations.
"Optimizing Indian Sign Language Detection via YOLOv8 Classification with SE Block Integration and MediaPipe Image Processing"	Jan 2026 / ResearchGate / IEEE	Reported 8–15% accuracy drops under sudden webcam lighting changes; proposed spatial focus enhancements to reduce background overfitting.	Backed our 3x Roboflow data-augmentation pipeline; we applied their environmental sensitivity findings to select Blur (1.7px), Noise (2.67%), and Brightness (±25%) settings to improve robustness to bedroom/lab lighting.	Supports targeted 3x data augmentation to mitigate environment and lighting overfitting.
"Deep Learning for India's Diverse Sign Languages: A Comprehensive Review from CNNs to Transformers"	June 2026 / JCMM / ResearchGate	Analyzed data scarcity and "signer dependency bias" across Indian sub-languages (2020–2026), highlighting severe text-to-sign access gaps.	Informed our data-prep: recognized that small, unbalanced samples cause failure; implemented Hard Negative Mining and class equalization to remove favoritism toward characters like Kha and Ga.	Emphasizes strict class equalization and Hard Negative Mining to eliminate dataset bias.

MAPPING TO SUSTAINABLE DEVELOPMENT GOALS (SDGs)

2026

SDG 10: Reduced Inequalities
Mapping:
The proposed system directly empowers the deaf and speech-impaired community by bridging the regional digital communication divide.
It provides equal access to essential public services such as hospitals, banks, government offices, and public transportation without requiring a human sign language translator.



SDG 9: Industry, Innovation, and Infrastructure
Mapping:
The project introduces next-generation digital infrastructure by combining edge computing, Mobile WebRTC, cloud-based microservices, computer vision, and Agentic Artificial Intelligence.
The system demonstrates how emerging technologies can be used to develop scalable and socially impactful accessibility solutions.



SDG 3: Good Health and Well-Being
Mapping:
The system incorporates a dedicated SOS and Emergency Override feature.
This enables deaf and speech-impaired individuals to instantly communicate critical medical conditions and emergency requirements to healthcare professionals.



CURRENT CHALLENGES & MAJOR PROJECT TRANSITION

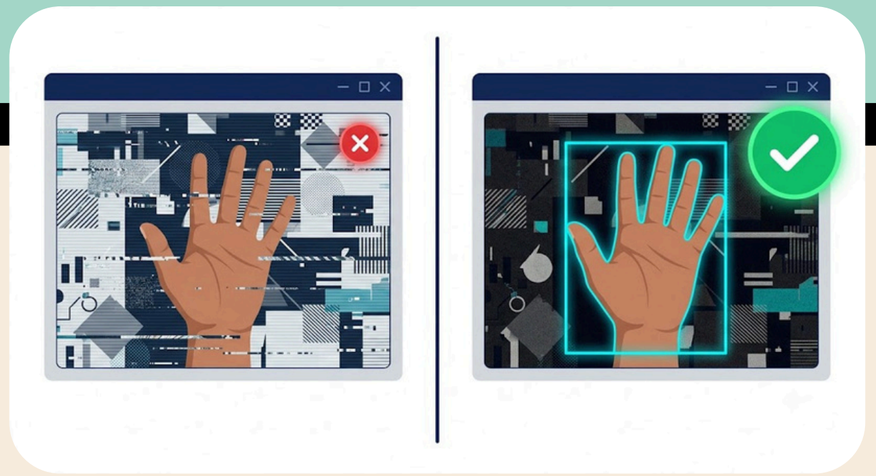
2026

Moving from Static Images to Real-Time Dynamic Processing

The Real-Time Challenge
The current Proof of Concept performs effectively on static and controlled datasets.
However, it faces the problem of Domain Shift when applied to real-time webcam environments.
Major challenges include dynamic lighting conditions, camera movement, complex backgrounds, environmental noise, and continuous hand movements.

Roboflow Dataset Pipeline – Under Progress
We are currently collecting, curating, annotating, and artificially augmenting a large dynamic Marathi Sign Language dataset using Roboflow.
The objective is to prepare the model for unpredictable and real-world camera environments.

YOLOv8 Integration – Work in Progress
We are actively transitioning from standard spatial filtering and CNN-based classification toward the YOLOv8-Nano object detection architecture.
The primary objective of the Final Year Major Project is to achieve real-time dynamic gesture recognition with high accuracy and efficient local processing.



PROPOSED METHODOLOGY – WHAT WE WILL BUILD

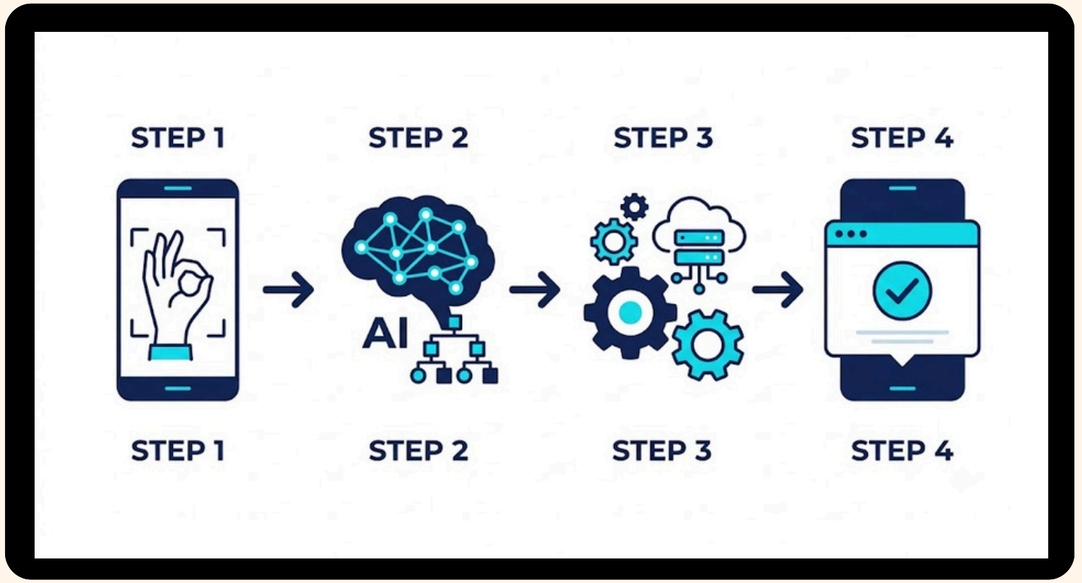
2026

Phase 1: Cloud-Native Microservices
The existing local Python-based Proof of Concept will be migrated to a scalable FastAPI backend server.
The system will receive and process live video frames through WebSockets and WebRTC communication.

Phase 2: Real-Time Gesture Recognition
YOLOv8-Nano will be trained on the custom dynamic Marathi Sign Language dataset.
The model will detect and classify hand gestures from continuous live video streams.

Phase 3: Agentic AI Integration
The existing Large Language Model will be upgraded from a basic sentence-generation system into an Action-Oriented AI Agent.
For example, if a user signs "I need a taxi," the AI Agent can understand the intent and interact with external services and APIs to assist the user.

Phase 4: Enterprise Communication Dashboard
A MERN Stack-based web portal will be developed for institutions such as hospitals, banks, and government offices.
The dashboard will receive live Sign-to-Text translations and allow hearing individuals to send Text-to-Sign responses.



Phase 1
FastAPI + WebRTC
→
Phase 2
YOLOv8 Training
→
Phase 3
Agentic AI + RAG
→
Phase 4
MERN Dashboard
→
Bi-Directional
Communication

PROJECT OBJECTIVES & EXPECTED OUTCOMES

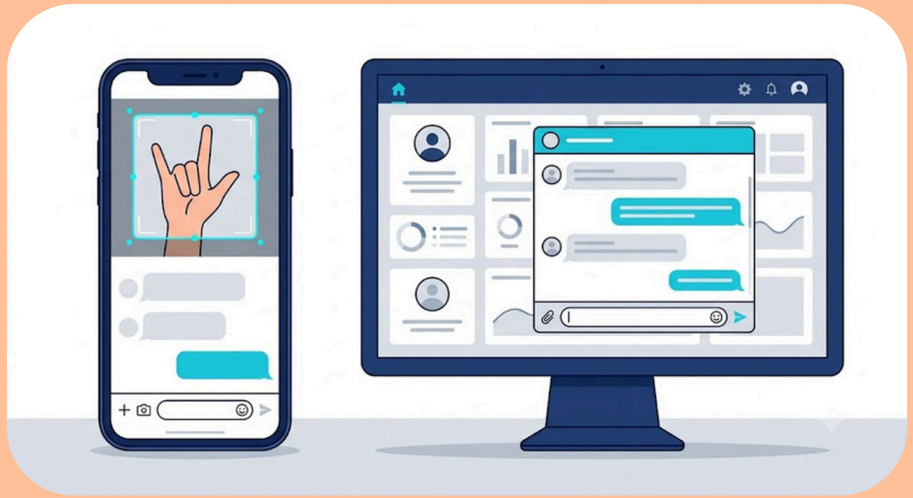
2026

Objectives

- Develop a robust, lighting-invariant Marathi Sign Language recognition model.
- Enable real-time dynamic gesture recognition from live video streams.
- Achieve end-to-end latency below 500 ms.
- Integrate Agentic AI + RAG for context-aware responses.
- Build a scalable bi-directional communication platform.
- Support Sign-to-Text, Sign-to-Speech, and Text-to-Sign translation.

Expected Outcomes

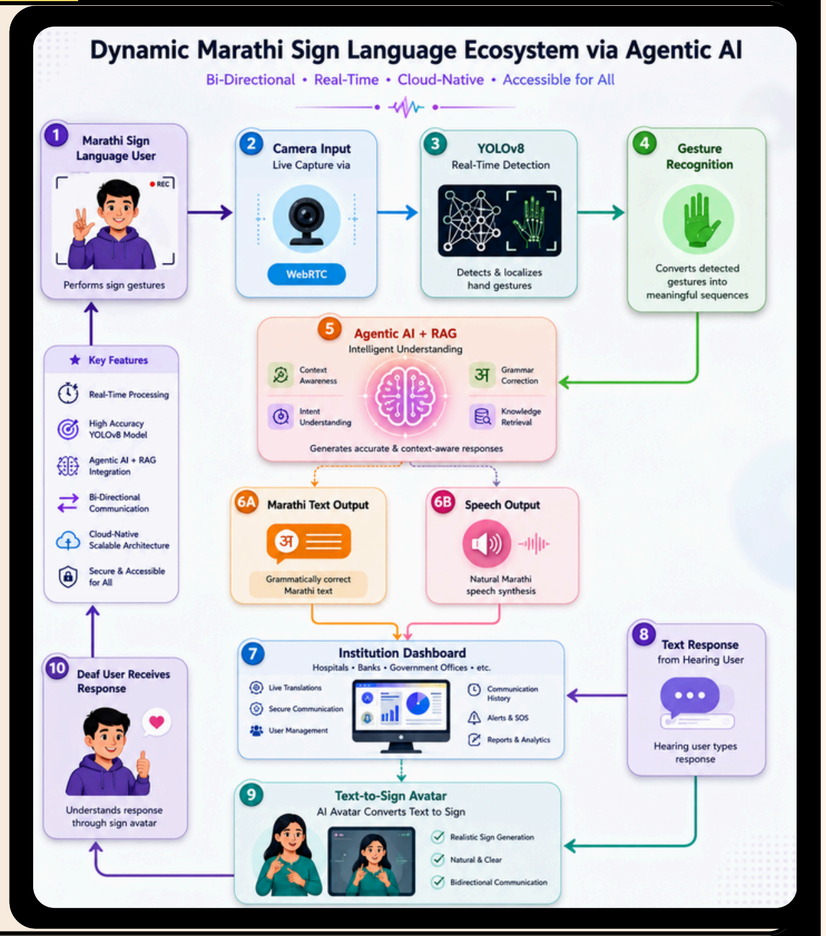
- Real-time Marathi Sign Language recognition system.
- Mobile application for hearing-impaired users.
- Secure institutional communication dashboard.
- Seamless two-way communication ecosystem.
- Agentic AI-powered intelligent assistance.
- Research publication in an IEEE/Springer conference.



Proposed System Architecture

Workflow

- Video Capture – User performs sign gestures via mobile/web camera.
- Data Streaming – Frames transmitted using WebRTC & WebSockets.
- YOLOv8 Detection – Real-time gesture recognition.
- Sequence Processing – Gestures combined into meaningful words.
- Agentic AI + RAG – Context understanding and response generation.
- Output Generation – Marathi Text & Speech.
- Enterprise Dashboard – Live communication for institutions.
- Text-to-Sign Avatar – Reverse communication for hearing users.





- Vision Layer: YOLOv8 + OpenCV
- AI Layer: Agentic AI + RAG + Llama3
- Application Layer: React Native + MERN Stack
- Infrastructure Layer: FastAPI + MongoDB + AWS



WebRTC / FastAPI
MongoDB + AWS

→

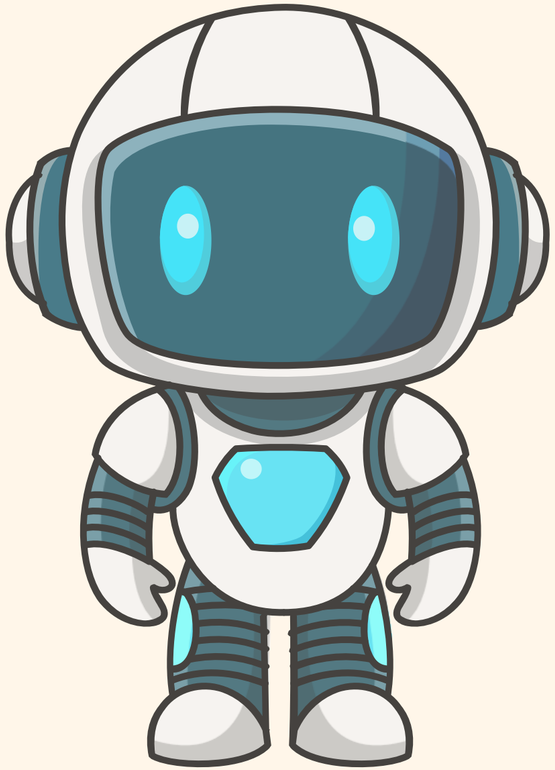
YOLOv8 + OpenCV
Gesture tracking

→

Agentic AI + RAG
LangChain + Llama3

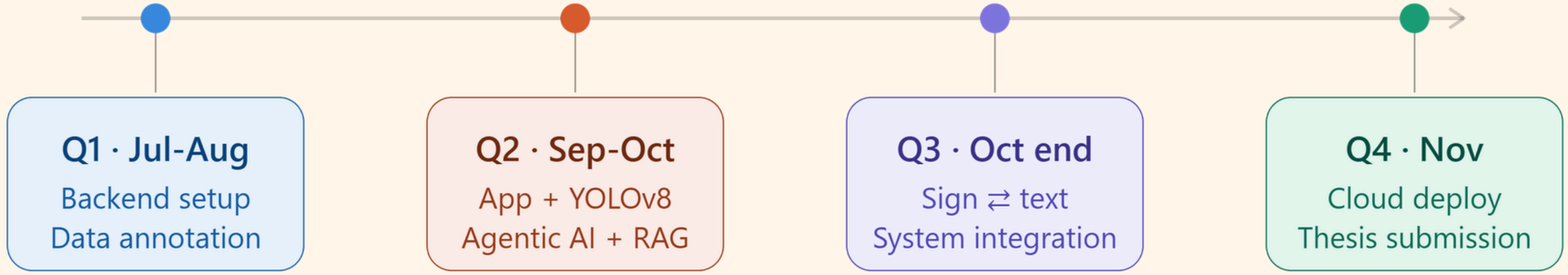
→

React Native
MERN dashboard



Timeline & Roadmap

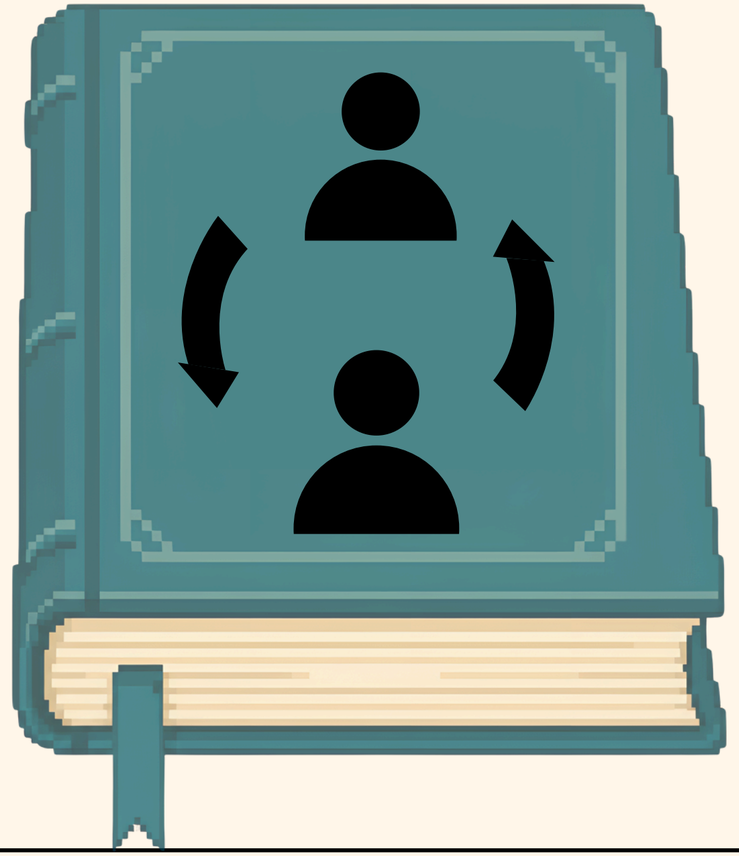
2026



- Q1 (Jul–Aug): Core backend laid down with FastAPI, MongoDB, and WebRTC, alongside dataset annotation for gesture training.
- Q2 (Sep–Oct): Mobile app development begins in React Native while YOLOv8 is trained; agentic AI and RAG pipelines get integrated.
- Q3 (Oct end): Dashboard completed, Sign-to-Text and Text-to-Sign modules built out, and all components brought together into one system.
- Q4 (Nov): Final push — cloud deployment, performance testing, and wrapping up the research paper and thesis submission.

Conclusion

- Extending our IVP Proof of Concept into a complete accessibility ecosystem.
- Combining YOLOv8, Agentic AI, RAG, FastAPI, React Native, and MERN Stack.
- Enabling real-time Marathi Sign Language communication.
- Delivering a scalable, intelligent, and market-ready accessibility platform.



THANK YOU!

Exploring the future of Image Processing together!